

JEEVAN JAYASURIYA

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SUMMARY

Ph.D. student in Industrial & Systems Engineering specializing in mechatronics, mechanical engineering, human factors, and biomechanics. Hands-on experience through research and prior industrial experience in embedded systems, CAD design, control systems, and prototyping. Seeking Summer 2026 industry internship opportunities.

EDUCATION

University of Wisconsin–Madison — Ph.D., Industrial & Systems Engineering (Expected May 2027)

Clarkson University — M.S., Mechanical Engineering, 2023

The Open University of Sri Lanka — B.Tech. (Hons.), Mechatronics Engineering, 2017

University of Sri Jayewardenepura — B.S. (Hons.), Physics, 2015

EXPERIENCE

UNIVERSITY OF WISCONSIN–MADISON

Madison, Wisconsin

Graduate Research Assistant – NeuroErgonomics Lab 2024–Present

- Collaborate with NASA Johnson Space Center to study effects of altered gravity, cognitive workload, and sleep deprivation on astronaut sensorimotor performance in human–machine systems.
- Design and execute long-duration human subject experiments (7–18 hrs) with synchronized multimodal data collection (LSL).
- Analyze multimodal time–series data (ambulatory brain imaging, EMG, eye tracking, kinematics, heart rate, force plates) using Temporal Convolutional Network (TCN)–based deep learning and statistical modeling.

CLARKSON UNIVERSITY

Potsdam, New York

Graduate Research Assistant – Astronautics and Robotics Lab 2022–2023

- Applied kinematics–driven musculoskeletal simulations to guide adaptive exoskeleton design for firefighter tasks, integrating biomechanics with mechatronics and mechanical engineering concepts.
- Gained hands-on experience in CAD design (SolidWorks), feedback control, microcontrollers, actuator and servo systems, sensor fusion, and system-level integration, along with prototyping, material selection, and 3D printing.

PRIOR PROFESSIONAL EXPERIENCE

- Lecturer – Embedded Systems, University of Sri Jayewardenepura (Sri Lanka)
- Scientific Officer – Radiation Protection, Sri Lanka Atomic Energy Board
- Assistant Manager – Corporate Banking, Peoples Bank
- Engineering Trainee – Robotics, Automation, PLC Systems

SELECTED PUBLICATIONS

- *Ergonomics* (2026): Firefighter motion capture & musculoskeletal simulation–driven biomechanical analysis ([article](#))
- *Int. J. of Coal Science & Technology* (2022): Machine vision-based automated fire-control system ([article](#))

TECHNICAL SKILLS

Sensing & Biomechanics: fNIRS (Homer3, MNE-Python), EMG, ECG/HR, eye tracking, Xsens, force plates, AnyBody

Data & ML: Python, PyTorch, TensorFlow, R, MATLAB, time-series analysis

Engineering Tools: Simulink, SolidWorks, PTC Creo, OpenCV, C++, Java, microcontrollers, embedded sensors & actuators

AWARDS & AFFILIATIONS

Best Student Work Award in the Aerospace Systems Technical Group (ASTG) in ASPIRE (2024 & 2025)

Academic Excellence Awards in Engineering (2016 & 2017)

Associate Member, Institute of Engineers Sri Lanka (IESL) (2018–present)

Student Affiliate Human Factors and Ergonomics Society (HFES) (2024–present)